```python
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.linear_model import LinearRegression
import joblib
import pickle
from google.colab import files

print("1. Waking up the data and training the AI...")
# Fetch dataset
url = "https://raw.githubusercontent.com/justmarkham/pycon-2016-tutorial/mas
df = pd.read_table(url, header=None, names=['label', 'clean_text'])
df['label'] = df['label'].map({'ham': 0, 'spam': 1})
df['clean_text'] = df['clean_text'].astype(str)

# Train the AI
X_train, X_test, y_train, y_test = train_test_split(df['clean_text'], df['la
vectorizer = TfidfVectorizer(max_features=5000)
X_train_multi = vectorizer.fit_transform(X_train)
reg_multi = LinearRegression().fit(X_train_multi, y_train)


print("\n2. Freeze-drying with JOBLIB (Optimized for Machine Learning)...")
joblib.dump(reg_multi, 'omniguard_model.joblib')
joblib.dump(vectorizer, 'omniguard_vectorizer.joblib')


print("3. Freeze-drying with PICKLE (Standard Python Method)...")
# Pickle requires you to "open" a file in "Write Binary" (wb) mode first
with open('omniguard_model.pkl', 'wb') as f:
    pickle.dump(reg_multi, f)

with open('omniguard_vectorizer.pkl', 'wb') as f:
    pickle.dump(vectorizer, f)


print("\n4. Downloading all 4 files straight to your computer...")
# Download Joblib files
files.download('omniguard_model.joblib')
files.download('omniguard_vectorizer.joblib')
# Download Pickle files
files.download('omniguard_model.pkl')
files.download('omniguard_vectorizer.pkl')

print("✅ Success! You now have both Joblib and Pickle versions.")
```

```
1. Waking up the data and training the AI...

2. Freeze-drying with JOBLIB (Optimized for Machine Learning)...
3. Freeze-drying with PICKLE (Standard Python Method)...

4. Downloading all 4 files straight to your computer...
✅ Success! You now have both Joblib and Pickle versions.
```

```
:-HOT ENCODING ---")

ly downloaded
om/justmarkham/pycon-2016-tutorial/master/data/sms.tsv"
ımes=['label', 'clean_text'])
), 'spam': 1})

:ked WHERE the message came from
;', or 'WhatsApp' to every message
 ]
[platforms) for _ in range(len(df))]

otice the words in platform_source):")
 clean_text']].head())

:==========================================

:==========================================
ource' column into 3 separate binary columns
;=['platform_source'], dtype=int)

ıdy for Machine Learning):")
ource_Email', 'platform_source_SMS', 'platform_source_WhatsApp']].head())
```

```
--- UPGRADING OMNIGUARD WITH ONE-HOT ENCODING ---

1. BEFORE ONE-HOT ENCODING (Notice the words in platform_source):
   label platform_source                                        clean_text
0      0           Email  Go until jurong point, crazy.. Available only ...
1      0        WhatsApp                        Ok lar... Joking wif u oni...
2      1        WhatsApp  Free entry in 2 a wkly comp to win FA Cup fina...
3      0           Email  U dun say so early hor... U c already then say...
4      0             SMS  Nah I don't think he goes to usf, he lives aro...

2. AFTER ONE-HOT ENCODING (Ready for Machine Learning):
   label  platform_source_Email  platform_source_SMS  platform_source_WhatsApp
0      0                      1                    0                         0
```